

#1. $m = 100 \text{ kg}$. $r = 6578 \text{ km} \rightarrow \omega = \sqrt{228600/6578^3} \approx 1.1834 \times 10^{-3} \text{ /sec.}$

a. Let $\vec{r} = [x, y, z]^T$ and $\vec{F} = [F_x, F_y, F_z]^T$.

Then $m\ddot{\vec{r}} = \vec{F} \Rightarrow \ddot{\vec{r}} = \vec{F}/m$.

$$\Rightarrow \dot{\vec{r}} = (\vec{F}/m)t + \dot{\vec{r}}_0$$

$$\Rightarrow \vec{r} = \frac{1}{2}(\vec{F}/m)t^2 + \dot{\vec{r}}_0 t + \vec{r}_0$$

$$\Rightarrow \begin{cases} x(t) = \frac{1}{2}(F_x/m)t^2 + \dot{x}_0 t + x_0 \\ y(t) = \frac{1}{2}(F_y/m)t^2 + \dot{y}_0 t + y_0 \\ z(t) = \frac{1}{2}(F_z/m)t^2 + \dot{z}_0 t + z_0 \end{cases}$$

If $\vec{F} = 0$, then $\begin{cases} x(t) = \dot{x}_0 t + x_0 \\ y(t) = \dot{y}_0 t + y_0 \\ z(t) = \dot{z}_0 t + z_0 \end{cases} \dots (A)$

Now consider the analytic solutions of CW equations.

If $\omega t \ll 1$, $\sin \omega t \approx \omega t$ and $\cos \omega t \approx 1$.

$$\Rightarrow \begin{cases} x(t) = (\dot{x}_0/\omega)\omega t - (3x_0 + 2\dot{y}_0/\omega) + 4x_0 + 2\dot{y}_0 t \omega \\ y(t) = (6x_0 + 4\dot{y}_0/\omega)\omega t + (2\dot{x}_0/\omega) - (6x_0 + 3\dot{y}_0)t + y_0 - 2\dot{x}_0 t \omega \\ z(t) = (\dot{z}_0/\omega)\omega t + z_0 \end{cases}$$

$$\Rightarrow \begin{cases} x(t) = \dot{x}_0 t + x_0 \\ y(t) = \dot{y}_0 t + y_0 \\ z(t) = \dot{z}_0 t + z_0 \end{cases} \text{ which is same as (A)}$$

Thus, the approximations of sine and cosine should be within accuracy of about 3%.

$$\begin{cases} \omega t - \sin \omega t \leq 0.03 \sin \omega t \\ 1 - \cos \omega t \leq 0.03 \cos \omega t \end{cases} \Rightarrow \begin{cases} \omega t \leq 1.03 \sin \omega t \dots (B) \\ 1 \leq 1.03 \cos \omega t \dots (C) \end{cases}$$

From (C), $(1.03)^{-1} \leq \cos \omega t \Rightarrow \omega t \leq 0.2419$.

Since $1.03 \sin(0.2419) = 0.2468 \geq 0.2419$, then $\omega t \leq 0.2419$ is satisfying both (B) and (C).

Also, for satisfying approximated formula with in accuracy of about 3%, ωt should be less than or equal to 3% of period of trigonometry function $\Rightarrow \omega t \leq 0.03(2\pi) = 0.1885$.

Hence $\omega t \leq 0.1885 \Rightarrow t \leq 0.1885/\omega \approx 102.7784 \text{ sec.}$

Thus the maneuver time period is 102.7784 sec.

b. $\Delta \vec{r}(0) = -10\hat{z} \text{ km}$. $\Delta \dot{\vec{r}}(0) = 0$. $\Delta \vec{r}(60) = 0 \text{ km}$, $\Delta \dot{\vec{r}}(60) = 0$.

$t=0$. $t=0^+ \rightarrow t=60^-$ $t=60$.

$$\begin{pmatrix} \Delta \vec{r}(0) = -10\hat{z} \\ \Delta \dot{\vec{r}}(0) = 0 \end{pmatrix} \begin{pmatrix} \Delta \vec{r}(0) = 0 \\ \Delta \dot{\vec{r}}(0) = \Delta \vec{V} \end{pmatrix} \begin{pmatrix} \Delta \vec{r}(60^-) = \Delta \vec{r}(0) \cdot 60 + \Delta \dot{\vec{r}}(0) \\ \Delta \dot{\vec{r}}(60^-) = \Delta \vec{V} \end{pmatrix} \begin{pmatrix} \Delta \vec{r}(60) = 0 \\ \Delta \dot{\vec{r}}(60) = 0 \end{pmatrix}$$

$$\Rightarrow \Delta \vec{r}(0) \cdot 60 + \Delta \vec{r}(0) = 0$$

$$\Rightarrow \Delta \vec{V} \cdot 60 - 10\hat{z} = 0$$

$$\Rightarrow \Delta \vec{V} = \frac{1}{6}\hat{z}$$

Thus, the initial $\Delta \vec{V}_i = \Delta \vec{V} = (1/6)\hat{z}$.

the final $\Delta \vec{V}_f = -\Delta \vec{V} = (-1/6)\hat{z}$.

#2. $L_2: (\bar{x}, \bar{y}, \bar{z}) = (-444645.6, 0, 0) \text{ km}$.

a. Let $x = \bar{x} + \Delta x$, $y = \Delta y$, $z = \Delta z$.

$$\begin{aligned} r_E^3 &= ((x - D_E)^2 + y^2 + z^2)^{-3/2} \\ &= ((\bar{x} + \Delta x - D_E)^2 + \Delta y^2 + \Delta z^2)^{-3/2} \\ &= ((\bar{x} - D_E)^2 + 2\Delta x(\bar{x} - D_E) + \Delta x^2)^{-3/2} \\ &= ((\bar{x} - D_E)^2 + 2\Delta x(\bar{x} - D_E))^{-3/2} \\ &= |\bar{x} - D_E|^{-3} \left(1 + 2\Delta x / (\bar{x} - D_E)\right)^{-3/2} \quad \left(\because \left|\frac{\Delta x}{\bar{x} - D_E}\right| \ll 1\right) \\ &= |\bar{x} - D_E|^{-3} \left(1 - 3\Delta x / (\bar{x} - D_E)\right) \end{aligned}$$

$$\begin{aligned} r_M^3 &= ((x + D_M)^2 + y^2 + z^2)^{-3/2} \\ &= ((\bar{x} + \Delta x + D_M)^2 + \Delta y^2 + \Delta z^2)^{-3/2} \\ &= ((\bar{x} + D_M)^2 + 2\Delta x(\bar{x} + D_M) + \Delta x^2)^{-3/2} \\ &= (\bar{x} + D_M)^{-3} \left(1 + 2\Delta x / (\bar{x} + D_M)\right)^{-3/2} \quad \left(\because \left|\frac{\Delta x}{\bar{x} + D_M}\right| \ll 1\right) \\ &= |\bar{x} + D_M|^{-3} \left(1 - 3\Delta x / (\bar{x} + D_M)\right) \end{aligned}$$

① $\ddot{x} - 2\omega_0 \dot{y} - \omega_0^2 x = -\frac{\mu_E(x - D_E)}{r_E^3} - \frac{\mu_M(x + D_E)}{r_M^3} + \frac{F_x}{m}$

Left side : $\ddot{x} + \Delta \ddot{x} - 2\omega_0(\dot{y} + \Delta \dot{y}) - \omega_0^2(\bar{x} + \Delta x)$
 $= \Delta \ddot{x} - 2\omega_0 \Delta \dot{y} - \omega_0^2 \Delta x - \omega_0^2 \bar{x}$.

Right side : $-\frac{\mu_E(\bar{x} + \Delta x - D_E)}{r_E^3} - \frac{\mu_M(\bar{x} + \Delta x + D_E)}{r_M^3} + \frac{F_x}{m}$
 $= -\frac{\mu_E(\bar{x} - D_E + \Delta x)}{|\bar{x} - D_E|^3} \left(1 - 3\frac{\Delta x}{\bar{x} - D_E}\right) - \frac{\mu_M(\bar{x} + D_E + \Delta x)}{|\bar{x} + D_M|^3} \left(1 - 3\frac{\Delta x}{\bar{x} + D_M}\right) + \frac{F_x}{m}$
 $= -\frac{\mu_E(\bar{x} - D_E)}{|\bar{x} - D_E|^3} - \frac{4\mu_E \Delta x}{|\bar{x} - D_E|^3} - \frac{\mu_M(\bar{x} + D_M)}{|\bar{x} + D_M|^3} - \frac{4\mu_M \Delta x}{|\bar{x} + D_M|^3} + \frac{F_x}{m}$
 $= -\omega_0^2 \bar{x} - 4\omega_0^2 \Delta x + F_x/m$.

$\Rightarrow \Delta \ddot{x} - 2\omega_0 \Delta \dot{y} - \omega_0^2 \Delta x - \omega_0^2 \bar{x} = -\omega_0^2 \bar{x} - 4\omega_0^2 \Delta x + F_x/m$.

$\Rightarrow \Delta \ddot{x} - 2\omega_0 \Delta \dot{y} + 3\omega_0^2 \Delta x = F_x/m$.

② $\ddot{y} + 2\omega_0 \dot{x} - \omega_0^2 y = -\frac{\mu_E y}{r_E^3} - \frac{\mu_M y}{r_M^3} + \frac{F_y}{m}$

Left side : $\ddot{y} + 2\omega_0(\dot{x} + \Delta \dot{x}) - \omega_0^2(\Delta y) = \ddot{y} + 2\omega_0 \Delta \dot{x} - \omega_0^2 \Delta y$.

Right side : $-\frac{\mu_E y}{r_E^3} - \frac{\mu_M y}{r_M^3} + \frac{F_y}{m}$
 $= -\frac{\mu_E \Delta y}{|\bar{x} - D_E|^3} \left(1 - 3\frac{\Delta x}{\bar{x} - D_E}\right) - \frac{\mu_M \Delta y}{|\bar{x} + D_M|^3} \left(1 - 3\frac{\Delta x}{\bar{x} + D_M}\right) + \frac{F_y}{m}$
 $= -\frac{\mu_E \Delta y}{|\bar{x} - D_E|^3} - \frac{\mu_M \Delta y}{|\bar{x} + D_M|^3} + \frac{F_y}{m} = -\omega_0^2 \Delta y + \frac{F_y}{m}$.

$\Rightarrow \ddot{y} + 2\omega_0 \Delta \dot{x} - \omega_0^2 \Delta y = \omega_0^2 \Delta y + F_y/m$.

$\Rightarrow \ddot{y} + 2\omega_0 \Delta \dot{x} = F_y/m$.

③ $\ddot{z} = -\frac{\mu_E z}{r_E^3} - \frac{\mu_M z}{r_M^3} + \frac{F_z}{m}$

Left side : $\ddot{z}$

Right side $-\frac{\mu_E z}{r_E^3} - \frac{\mu_M z}{r_M^3} + \frac{F_z}{m}$

$$= -\frac{\mu_E \Delta z}{|\vec{r}-\vec{r}_E|^3} \left\{ 1 - 3 \frac{\vec{r} \cdot \vec{r}_E}{|\vec{r}-\vec{r}_E|^2} \right\} - \frac{\mu_M \Delta z}{|\vec{r}+\vec{r}_M|^3} \left\{ 1 - 3 \frac{\vec{r} \cdot \vec{r}_M}{|\vec{r}+\vec{r}_M|^2} \right\} + \frac{F_z}{m}$$

$$= -\frac{\mu_E \Delta z}{|\vec{r}-\vec{r}_E|^3} - \frac{\mu_M \Delta z}{|\vec{r}+\vec{r}_M|^3} + \frac{F_z}{m} = -\omega_0^2 z + \frac{F_z}{m}$$

$$\Rightarrow \ddot{\Delta z} + \omega_0^2 \Delta z = F_z/m \dots (A)$$

b. From (A), $\begin{bmatrix} \ddot{\Delta z} \\ \dot{\Delta z} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\omega_0^2 & 0 \end{bmatrix} \begin{bmatrix} \Delta z \\ \dot{\Delta z} \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} \frac{F_z}{m}$

Characteristic eqn, $\lambda^2 + \omega_0^2 = 0$

$\therefore$ eigenvalues : $\lambda = \pm i\omega_0$

c. Since eigenvalues are pure imaginary $\#$, then out of plane motion is marginally stable.

#3 $H = \int_B \vec{F}(m) \times \vec{v}(m) dm$

$$= \int_B \vec{F} \times (\vec{\omega} \times \vec{r}) dm$$

$$= \int_B (\vec{r}^T \vec{F} - \vec{F}^T \vec{r}) \vec{\omega} dm$$

$$= \int_B \begin{bmatrix} r_y^2 + r_z^2 & -r_x r_y & -r_x r_z \\ -r_x r_y & r_x^2 + r_z^2 & -r_y r_z \\ -r_x r_z & -r_y r_z & r_x^2 + r_y^2 \end{bmatrix} dm \vec{\omega} = \vec{I} \vec{\omega}$$

By Coriolis's thm,

$$\left(\frac{dH}{dt} \right)_{\text{inertial}} = \left(\frac{dH}{dt} \right)_{\text{body}} + \vec{\omega} \times H$$

$$\Rightarrow \dot{H} = (\vec{I} \dot{\vec{\omega}})' + \vec{\omega} \times H$$

$$\Rightarrow \vec{I} \dot{\vec{\omega}} = \vec{\omega} \times \vec{I} \vec{\omega}$$

$$\therefore \vec{I} \dot{\vec{\omega}} = \vec{\omega} \times \vec{I} \vec{\omega} + \vec{M}$$

Since $\vec{I} = \text{diag}(A, B, C)$, then

$$\begin{cases} A \dot{\omega}_1 = (B-C) \omega_2 \omega_3 + M_1 \\ B \dot{\omega}_2 = (C-A) \omega_1 \omega_3 + M_2 \\ C \dot{\omega}_3 = (A-B) \omega_1 \omega_2 + M_3 \end{cases}$$

b. Let $T = \frac{1}{2} \omega^T \vec{I} \omega$ which is the kinetic energy.

$$\frac{dT}{dt} = \frac{d}{dt} \left(\frac{1}{2} \omega^T \vec{I} \omega \right) = \omega^T \dot{\vec{I}} \omega = \omega^T (\vec{\omega} \times \vec{I} \vec{\omega}) = \vec{\omega} \cdot (\vec{\omega} \times \vec{I} \vec{\omega})$$

$$= \det \begin{pmatrix} \omega_1 & \omega_2 & \omega_3 \\ \omega_1 & \omega_2 & \omega_3 \\ A\omega_1 & B\omega_2 & C\omega_3 \end{pmatrix} = 0 \therefore T \text{ is conserved.}$$

Let $H = |\vec{H}| \neq 0$.

$$\frac{d}{dt} H^2 = 2\vec{H} \cdot \dot{\vec{H}} = 2(\vec{I}\vec{\omega})^T (\dot{\vec{I}}\vec{\omega}) = 2(\vec{I}\vec{\omega})^T (\vec{\omega} \times \vec{I}\vec{\omega})$$

$$= \det \begin{bmatrix} A\omega_1 & B\omega_2 & C\omega_3 \\ \dot{A}\omega_1 & \dot{B}\omega_2 & \dot{C}\omega_3 \end{bmatrix} = 0.$$

Thus the magnitude of angular momentum H is conserved.

#Q. Same as other Final exam.